Unitary Matrix

Unitary Matrix is a square matrix of complex numbers. The product of the conjugate transpose of a unitary matrix, with the unitary matrix, give

Let us learn more about the properties, and examples of the unitary matrix.

What is Unitary Matrix?

A **unitary matrix** is a square matrix of complex numbers, whose inverse is equal to its conjugate transpose. Alternatively, the product of the uni

- $U^H = U^{-1}$
- $U^H U = U U^H = I$

where I is the identity matrix whose order is the same as U.

Also, a unitary matrix is a nonsingular matrix. Or the determinant of a unitary matrix is not equal to zero. The columns and rows of a unitary mat

Properties of Unitary Matrix

The properties of a unitary matrix are as follows.

- The unitary matrix is a non-singular matrix.
- The unitary matrix is an invertible matrix
- The product of two unitary matrices is a unitary matrix.
- The inverse of a unitary matrix is another unitary matrix.
- A matrix is unitary, if and only if its transpose is unitary.
- A matrix is unitary if its rows are orthonormal, and the columns are orthonormal.
- The unitary matrices can also be non-square matrices but have orthonormal columns and rows.

Note: The sum or difference of two unitary matrices doesn NOT need to be a unitary matrix. For example, if A is a unitary matrix, then A - A = O

Terms Related to Unitary Matrix

The following terms related to matrices are helpful for a better understanding of this concept of unitary matrix.

- **Non-Singular Matrix:**The determinant of a non singular matrix is a a non zero value. For a square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the condition of it bei
- **Invertible Matrix:** The matrix whose inverse matrix can be computed, is called an invertible matrix. The inverse of a matrix A is A^{-1} = Adj A/
- **Conjugate Matrix:** The conjugate matrix of a given matrix is obtained by replacing the corresponding elements of the given matrix, with the
- **Transpose Matrix:** The transpose of a matrix A is represented as A^T, and the transpose of a matrix is obtained by changing the rows into co
- **Orthogonal Matrix:** If the product of a matrix and its transpose is an identity matrix, then it is called an orthogonal matrix. $A.A^T = I$.
- **Hermitian Matrix:** A hermitian matrix is a square matrix, which is equal to its conjugate transpose matrix. The non-diagonal elements of a h

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Examples on Unitary Matrix

Example 1: Show that the matrix A = $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1i}{\sqrt{2}} & \frac{-1i}{\sqrt{2}} \end{bmatrix}$ is a unitary matrix.

Solution:

The given matrix is A = $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1i}{\sqrt{2}} & \frac{-1i}{\sqrt{2}} \end{bmatrix}$

Conjugate of matrix A = $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1i}{\sqrt{2}} & \frac{1i}{\sqrt{2}} \end{bmatrix}$

Practice Questions on Unitary Matrix

- $U^T = U^{-1}$
- $U^H = U^{-1}$
- $U^H = U$

FAQs on Unitary Matrix

What is the Definition of a Unitary Matrix?

How to Find the Complex Transpose Matrix?

How Do You Know If a Matrix is Unitary Matrix?

What Are the Properties of Unitary Matrix?

is a Unitary Matrix Also a Hermitian Matrix?

What is the Order of a Unitary Matrix?